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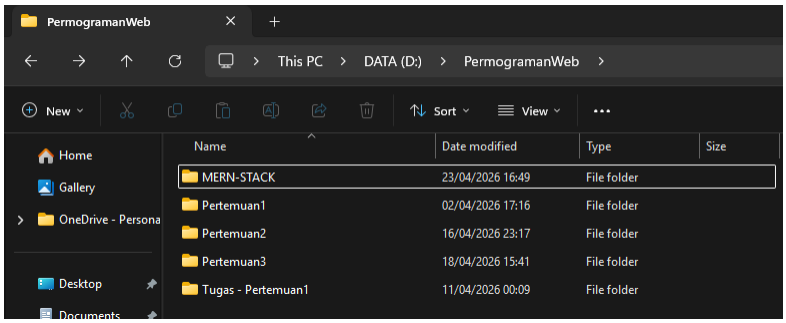
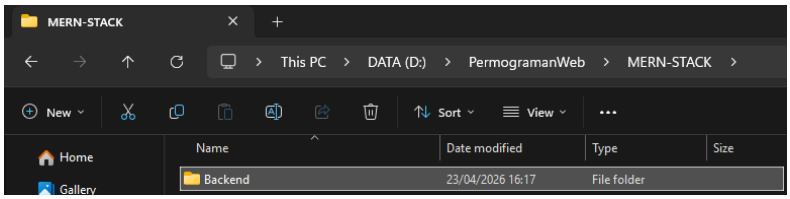
Matkul : Lab. Pemograman Web

TUGAS PERTEMUAN 4

Pertama kita buat folder MERN-STACK



Setelah itu buat folder Backend

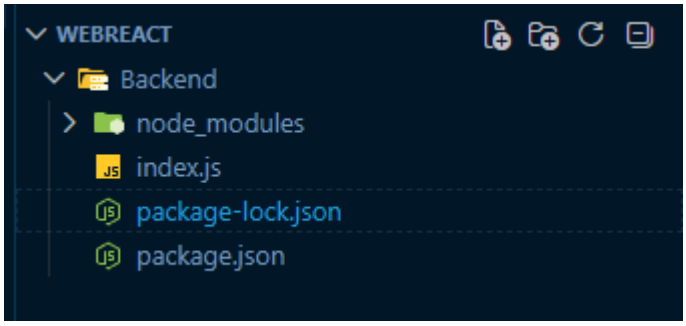


Lalu masuk ke visual studio code dan buat file backend serta tambahkan index.js pada folder Backend

```
D:\Documents\My VsCode Project\WebReact>cd Backend
```

```
D:\Documents\My VsCode Project\WebReact\Backend>npm init -y
Wrote to D:\Documents\My VsCode Project\WebReact\Backend\package.json:

{
  "name": "backend",
  "version": "1.0.0",
  "description": "",
  "main": "index.js",
  "scripts": {
    "test": "echo \"Error: no test specified\" && exit 1"
  },
  "keywords": [],
  "author": "",
  "license": "ISC",
  "type": "commonjs"
}
```

```
D:\Documents\My VsCode Project\WebReact\Backend>npm install express mysql2 cors

added 78 packages, and audited 79 packages in 5s

25 packages are looking for funding
  run `npm fund` for details

found 0 vulnerabilities
```



Di dalam file index.js kita tulis code seperti dibawah:

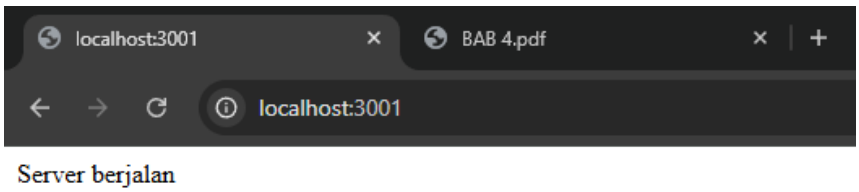
```
index.js   package.json
Backend > index.js > ...
1  const express = require("express");
2  const mysql = require("mysql2");
3  const cors = require("cors");
4
5  const app = express();
6
7  app.use (
8      cors ({
9          origin: "http://localhost:3000",
10         credentials: true,
11     }),
12 );
13
14 app.use(express.json());
15
16 const db = mysql.createConnection({
17   host: "localhost",
18   user: "root",
19   password: "",
20   database: "pw_praktikum_db",
21 });
22
23 db.connect((err) => {
24   if (err) {
25     console.error("Koneksi ke database gagal:", err.message);
26     process.exit(1);
27   }
28   console.log("Koneksi ke database MySQL berhasil!");
```

Kita memanggil server frontend melalui localhost: 3000, lalu menggunakan database MySql pw_praktikum_db

```
31  app.get("/", (req, res) => {
32    res.send("Server berjalan");
33  });
34
35  app.listen(3001, () => {
36    console.log("Server running on port 3001");
37  });
38
39  app.get("/api/info", (req, res) => {
40      res.json ({
41          message: "API MERN Stack Build by Express JS",
42          version: "1.0.0",
43          status: "active",
44          timestamp: new Date(),
45      });
46  });
```

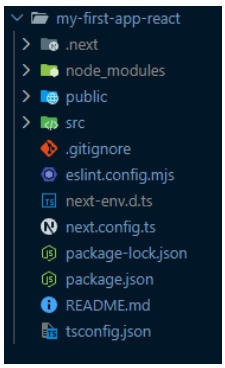
Lalu app.listen adalah code yg membuat file backend menggunakan localhost: 3001. Setelah itu jalankan server backend dengan mengetik “node index” pada termin

```
D:\Documents\My VsCode Project\WebReact\Backend>node index
Server running on port 3001
Koneksi ke database MySQL berhasil!
```



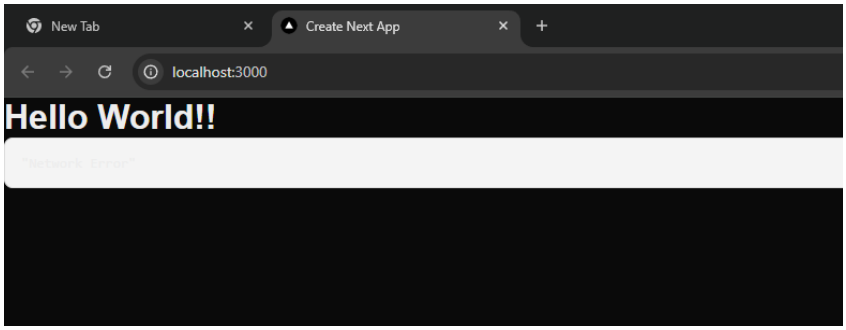
Selanjutnya buat folder frontend menggunakan “npx create-next-app@latest” pada terminal baru:

```
D:\Documents\My VsCode Project\WebReact>npx create-next-app@latest
√ What is your project named? ... my-first-app-react
√ Would you like to use the recommended Next.js defaults? » No, customize settings
√ Would you like to use TypeScript? ... No / Yes
√ Which linter would you like to use? » ESLint
√ Would you like to use React Compiler? ... No / Yes
√ Would you like to use Tailwind CSS? ... No / Yes
√ Would you like your code inside a `src/` directory? ... No / Yes
√ Would you like to use App Router? (recommended) ... No / Yes
√ Would you like to customize the import alias (`@/*` by default)? ... No / Yes
√ What import alias would you like configured? ... @/*
√ Would you like to include AGENTS.md to guide coding agents to write up-to-date Next.js code? ... No / Yes
Creating a new Next.js app in D:\Documents\My VsCode Project\WebReact\my-first-app-react.

Using npm.

Initializing project with template: app
```



Lalu pada src/app/page.tsx kita ubah code nya menjadi seperti ini:

```
index.js   page.tsx M ×   package.json M
my-first-app-react > src > app > page.tsx > Home
1  "use client";
2
3  import axios from "axios";
4  import { useEffect, useState } from "react";
5
6  export default function Home() {
7    const [info, setInfo] = useState<any>(null);
8
9    const getAPIInfo = async () => {
10     try {
11       const response = await axios.get("http://localhost:3001/api/info");
12       setInfo(response.data);
13     } catch (error: any) {
14       console.error("Error fetching API data:", error);
15       setInfo(error?.message);
16     }
17   };
18
19   useEffect(() => {
20     getAPIInfo();
21   }, []);
22
23   return (
24     <div>
25       <h1>Hello World!!</h1>
26
27       {info && (
28         <pre
29           style={{
30             backgroundColor: "#f4f4f4",
31             padding: "15px",
32             borderRadius: "8px",
33             border: "1px solid #ddd",
34             overflow: "auto",
35           }}
36         >
37           {JSON.stringify(info, null, 2)}
38         </pre>
39       )}
40     </div>
41   );
42 }
43
```

Pada frontend, kita mengambil server backend 3001 lalu setelah itu kita panggil menggunakan “npm run dev”

```
D:\Documents\My VsCode Project\WebReact\my-first-app-react>npm run dev

> my-first-app-react@0.1.0 dev
> next dev

▲ Next.js 16.2.4 (Turbopack)
- Local:         http://localhost:3000
- Network:       http://192.168.56.1:3000
✓ Ready in 486ms
```